MELLOX AI

05

Railway Deployment & Operations

Reproducible deployment, configuration, scheduled work, security boundaries, monitoring and recovery.

REBUILD BLUEPRINT SERIES / VOLUME 05 OF 06

Source snapshot: 21 October 2026 | Branch: master

Commit: 055c90977c2c | Internal working reference

Code-derived implementation. Production activation requires separate evidence.

01 How to use this volume

This is volume 05 of six documents describing Mellox AI as represented in the local source snapshot on 21 October 2026. The owner confirmed that this repository is current, Railway is the hosting platform, codebase pricing should be used, and mellox.ai is the public website.

Evidence labels: Implemented means source exists and is mapped here; it does not prove that a production migration, credential, permission or feature switch is enabled. Owner-confirmed identifies facts supplied by the owner. External-verified identifies official web information checked on the snapshot date. Recommended identifies a rebuild or operating procedure. Unverified identifies missing deployment or commercial evidence.

Runtime code, migrations and current configuration outrank older guides. Model and price tables describe defaults unless explicitly stated otherwise. They are a dated reference, not a promise that vendors will retain them. No credentials or customer records are included.

Volume	Read it for
01 Product & workflows	Feature behavior, journeys, approvals and module ownership
02 Architecture & engineering	Frontend, backend, integrations, APIs and implementation patterns
03 AI models & generation	Every task-model route, images, video, context and evaluations
04 Pricing & billing	Plans, feature gates, all action prices, wallets and economics
05 Railway & operations	Deployment, configuration, jobs, security, recovery and diagnostics
06 Rebuild & data reference	Build order, acceptance criteria, full typed schema and migration index

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02 Confirmed hosting and deployment boundary

Railway is owner-confirmed as the current host. The repository's root Dockerfile builds the main Next.js app; it does not by itself provision Supabase, R2, third-party OAuth apps, payment webhooks, schedules or optional Python/SDR services. Exact Railway project IDs, services, regions, replicas, resource sizes and deployed commit were not available to this audit.

Use this volume as a source-based operating recipe, not a claim that all steps are already applied to production. Existing Amplify configuration and older deployment ADRs remain repository artifacts, but are not adopted here as the production runbook. Do not redeploy to those platforms just because an old guide names them.

A deployable release is a compatible application build, database migration state, environment configuration, provider permissions, callback configuration, worker/scheduler availability and a recoverable previous release. Record all of those together. A green container build is not sufficient evidence that generation, payment, media access or scheduled delivery works.

Implementation sources: Dockerfile; next.config.ts; src/server/env.ts; docs/deployment-guide.md

03 Container build and runtime requirements

The Dockerfile uses Node 22 on Debian bookworm-slim in dependency, builder and runner stages. It installs dependencies from package-lock, builds Next.js, verifies the Clarity build, then copies standalone output, public assets, static chunks and vendor resources into the runtime image. The runtime uses a non-root nextjs user.

The final image includes Chromium and fonts for visual capture/render fallback, Python and ExifTool for image metadata, and FFmpeg for video assembly. GEO_RENDER_EXECUTABLE and VISUAL_CAPTURE_BROWSER_PATH point to the installed Chromium. If using a different Railway build strategy, reproduce those native runtime dependencies; package.json alone does not install them.

The process binds to HOSTNAME=0.0.0.0 and Railway's injected PORT. The Docker health command checks /health on that port. /api/health exposes safe capability and build identity; /api/health/ready is a deeper readiness surface. /health is the simple process endpoint and must not be mistaken for provider or database readiness.

Public NEXT_PUBLIC values are build inputs and can be embedded in client bundles. Provider and service-role secrets are runtime server values. Never place secret keys in build-public variables merely to make a browser feature connect. Changing a build-public URL generally requires rebuilding the client artifact.

Implementation sources: Dockerfile; src/app/health/route.ts; src/app/api/health/route.ts; src/app/api/health/ready/route.ts; scripts/start-production.mjs

04 Recommended fresh-environment deployment sequence

Step	Action	Exit evidence
1. Freeze release	Record commit, lockfile, migration list and provider policy.	Release manifest with approved environment target.
2. Provision state	Create isolated Supabase project, private R2 bucket and optional Redis.	Correct project IDs and access boundaries, without exposing secrets.
3. Apply schema	Verify and apply the canonical migration sequence to the new target.	Recorded migration history and regenerated/checked types.
4. Configure secrets	Set required core runtime values, encryption roots, provider keys and callback origins.	Production env validation and deliberate optional-feature readiness.
5. Build on Railway	Use root Dockerfile or equivalent dependency-complete build.	Build succeeds; standalone assets and native tools present.
6. Verify process	Bind PORT; configure Railway health path and custom HTTPS origin.	/health responds, deployed identity matches expected commit.
7. Wire providers	Set exact OAuth redirects, signed webhook endpoints and intended scopes.	Test account connection and webhook validation.
8. Enable schedules	Verify database/network cron target and secret; check heartbeat.	Due job advances exactly once without opening public writes.
9. Smoke test	Use a controlled workspace and permitted provider operations.	Auth isolation, content, media, balance and delivery evidence.
10. Release gates	Enable features deliberately, observe errors/cost and retain rollback.	Stable key paths plus recoverable previous release.

Implementation sources: Dockerfile; src/instrumentation.ts; src/server/env.ts; src/server/cron.ts; supabase/migrations; docs/storage.md

Repository validation commands

```
npm ci
npm run db:verify
npm run typecheck
npm run lint
npm test
npm run build
```

These are recommended release checks, not checks run during this read-only documentation review. Live suites need an isolated test workspace and explicit control over real provider calls and writes. npm run db:types regenerates checked-in types after an intentional schema change; migration application is a separate operation with the correct target credentials.

Railway healthchecks - Official documentation checked 21 October 2026. Use the current platform setting and injected port; confirm actual project configuration.
https://docs.railway.com/deployments/healthchecks

Railway variables - Official documentation checked 21 October 2026. Keep environment-specific variables and secret scopes deliberate.
https://docs.railway.com/variables

05 Required configuration and encryption continuity

Core server validation requires APP_URL, Supabase URL/publishable/service-role values, matching browser-safe Supabase values, OPENROUTER_API_KEY and CRON_SECRET. In production APP_URL must be the intended HTTPS origin rather than localhost. The catalog, callbacks, consent metadata and generated links depend on it. Refer to src/server/env.ts for current validation; some feature-specific modules validate additional variables outside that central schema.

R2 needs account ID, access key ID, secret access key and bucket name. An endpoint override is optional. A partially configured set fails explicitly rather than silently moving production media into another backend. The production storage path requires R2, while development can use legacy Supabase fallback.

Google, GitHub, WordPress, Webflow, Canva, Notion and Slack each have their own OAuth/app configuration. Google sign-in credentials are distinct from Google Analytics/Search Console authorization. Register exact callback origins and paths. Preserve connector encryption roots across releases; a rotated key without a migration can strand saved grants.

For distribution select Post for Me when intended and configure its key/webhook secret. SocialAPI is retired. SDR is optional and needs its own endpoint, tenant provisioning, secrets and workspace flag if selected. For billing, production defaults to enforcement on; payment keys, webhook secret, price mappings and schema readiness still need verification.

Implementation sources: src/server/env.ts; src/server/assets/storage.server.ts; src/server/connectors; src/server/billing/readiness.server.ts; src/lib/feature-flags.ts

06 Feature switches and readiness

Feature	Source default / prerequisite	Operational consequence
Autopilot	On unless false; plan and program setup still apply	Full automatic mode has its own flag; AGENTS_DISABLED/workspace pause can stop execution.
Audience	On unless false; separate auto-score flag	Worker reads existing shared scheduler; disabled mode hides/blocks feature paths.
Memory / chat tools	On unless false; workspace memory switch separately	Do not bypass editor/admin write rules.
MCP	Global gate on unless false; per-workspace access starts off	Admin enables access and separately permits changes.
Stories	On unless false, with workspace override	Requires Story schema plus account permissions; preserve frame recovery.
GEO scan probes	Explicit opt-in	Paid multi-engine checks; not the same switch as tracked-prompt allowances.
GEO rendering	Explicit configured fallback	Chromium and safe resource fetching required.
Proof Engine	Explicit opt-in	Plan and eligibility checks still apply.
Slack	Explicit opt-in	Provider installation and verified inbound binding required.
UGC	KIE prerequisite in current legacy helper plus feature gate	Runtime health/provider readiness is not identical to this helper; verify OpenRouter-only deployments carefully.
Canva Magic Layers	Preview flag should be false for public integration review	Flat-image fallback is a separate supported path.
Platform DB flags	Env override, then database rule, otherwise off	Fifteen-second cache; does not replace all older env gates.

Do not assume one universal flag system. Current code includes legacy environment gates, workspace overrides, account entitlements and newer database rollout flags. Platform flags, model-route overrides, spend caps and operational switches have separate services and caches. Confirm the gate used by the exact feature path.

Implementation sources: src/lib/feature-flags.ts; src/server/platform/flags.server.ts; src/server/platform/switches.server.ts; src/server/platform/spend-caps.server.ts; src/server/platform/routing-overrides.server.ts

07 Scheduled jobs and public hook map

These routes exist in current source. A cron hook is not enabled merely because its file exists: verify the scheduling migration, configured APP_URL/target origin, CRON_SECRET and heartbeat. Provider webhook routes require their signature protocol instead of the cron secret. Extracted interval values are code expectations, not a live scheduler inspection.

Hook	Job / expected interval seconds	Trigger source
/api/public/hooks/agents-tick	agents-tick / 15	Authenticated scheduler
/api/public/hooks/analytics-sync	analytics-sync / 600	Authenticated scheduler
/api/public/hooks/billing	billing / 300	Authenticated scheduler
/api/public/hooks/competitor-watch	competitor-watch / 30	Authenticated scheduler
/api/public/hooks/experiments	experiments / 300	Authenticated scheduler
/api/public/hooks/geo-agents	geo-agents / 60	Authenticated scheduler
/api/public/hooks/geo-scans	geo-scans / 60	Authenticated scheduler
/api/public/hooks/kie	provider/custom / see source	Provider webhook
/api/public/hooks/link-catalog	link-catalog / 86_400	Authenticated scheduler
/api/public/hooks/link-orders	link-orders / 60	Authenticated scheduler
/api/public/hooks/openrouter-video	provider/custom / see source	Provider webhook
/api/public/hooks/ops-watch	ops-watch / 300	Authenticated scheduler
/api/public/hooks/postforme	provider/custom / see source	Provider webhook
/api/public/hooks/run-schedules	run-schedules / 60	Authenticated scheduler
/api/public/hooks/sdr	provider/custom / see source	Provider webhook
/api/public/hooks/sdr-reconcile	sdr-reconcile / 300	Authenticated scheduler
/api/public/hooks/slack	slack / 60	Authenticated scheduler
/api/public/hooks/socialapi	provider/custom / see source	Provider webhook
/api/public/hooks/stripe	provider/custom / see source	Provider webhook
/api/public/hooks/ugc-renders	ugc-renders / 60	Authenticated scheduler

Implementation sources: src/app/api/public/hooks; src/server/cron.ts

Shared minute scheduler

run-schedules expects a sixty-second cadence. It runs up to twenty-five due scheduled jobs and twenty-five due Market Brain collections, Autopilot with a forty-five-second/max-twelve sweep, and Audience with a forty-second/max-six sweep. after() continuations refresh trends if due, collect audience outcomes, run bounded visual captures and purge expired memories. These continuations still require process time; deployments/restarts make durable recovery important.

Implementation sources: src/app/api/public/hooks/run-schedules/route.ts

08 Security boundaries and control matrix

Boundary	Implemented control	Verification to preserve
User identity	Supabase verified bearer/cookie session flows	Invalid recovery link cannot use an unrelated session.
Tenant access	Explicit workspace membership/role plus RLS	Outsider IDs and stale workspace context are rejected.
Service role	Server-only explicit privileged clients	Every privileged query is scoped and audited.
External fetch	DNS/private-address protection, redirect validation, size/time bounds	Block metadata/private hosts and redirect pivots.
Media	Workspace paths, signatures, prompt/image/video safety	Uploads and edits cannot bypass publish-time review.
OAuth	Bound one-use state, expiry, PKCE/cookie where implemented	Wrong browser/user/workspace and replay fail.
Webhooks	Raw-body signature/secret checks and replay-safe transitions	Invalid/stale messages cannot advance state.
Payment	Server pricing, holds, signed events, idempotent grants	Browser amount and success-page visit are not authority.
Repository writes	Verified source host, reviewed base and patch approval	No direct push/merge to protected base by a generator.
External assistants	Workspace gate, caller role, own app checks and audit	Unapproved content cannot be published via MCP.
Staff tools	Permissions, production MFA, recent MFA for sensitive actions	A customer admin is not automatically platform staff.

Implementation sources: src/server/route.ts; src/server/safe-fetch.ts; src/server/guardrails; src/server/mcp; src/server/admin/route.ts

09 Headers, browser sessions and privacy

next.config.ts sets CSP and security headers, API no-store behavior, frame restrictions, MIME-sniff prevention, referrer policy and production HSTS. The CSP retains unsafe-inline for required framework bootstrapping and JSON-LD; do not describe it as a strict per-request nonce policy. Allowed connect/frame hosts are deliberate integration requirements.

The Supabase browser client manages the current session flow, including legacy-session migration and explicit callback handling. Do not generalize an older localStorage description over the current cookie/SSR behavior. Client storage and UI guards do not replace server session verification.

Sensitive credential data belongs in server-only tables with encryption and minimal safe metadata exposed to clients. Helicone prompt-body logging is opt-in; telemetry configuration can affect data exposure. The repository does not establish a universal retention period, data residency guarantee, certified compliance status, signed data-processing contract or backup erasure guarantee. Those remain documented owner decisions, not assumptions.

Implementation sources: next.config.ts; src/integrations/supabase/client.ts; docs/security.md; docs/privacy.md; docs/self-hosted-integrations.md

10 Monitoring, readiness and incident triage

Start with /health for process liveness, /api/health for safe build/capability information and /api/health/ready for configured readiness. The health version uses Railway/Git commit metadata and process start time. A configured key is not proof that the vendor account is funded or an API request succeeds.

Useful durable signals include AI usage/reservations, billing holds and financial events, cron heartbeats, job status/lease age, webhook events, audit logs, provider failures, missing connections and queue lag. Optional Sentry, Helicone, Redis and alert sinks improve observability when configured. Track pending-job age and capture/release failures as well as HTTP error rate.

During an incident identify the affected release and workspace safely, check recent configuration/migration changes, isolate provider outage versus auth/credit/rate/safety failure, and disable the smallest affected feature path if necessary. Reconcile uncertain side effects from stored provider IDs and event history before retrying. Never repair a dashboard by manually marking an unknown post delivered or a GEO finding verified.

Implementation sources: src/app/api/health/route.ts; src/app/api/health/ready/route.ts; src/server/observability; src/server/cron.ts; docs/operations.md

Symptom	First checks	Safe recovery
App fails at boot	Required env validation, public build args, schema readiness	Restore correct config/release; keep encryption roots stable.
Jobs never advance	Heartbeat, scheduler target/secret, lease expiry, flags	Repair scheduler then bounded idempotent sweep.
Video stuck after deploy	Stored provider task, composition row, FFmpeg, poll route	Resume accepted tasks; do not submit blindly.
Media 403/404	Workspace path, storage backend marker, expired URL, R2 config	Re-sign through authorized route; preserve old objects.
Unexpected charge/balance	Hold/capture/event key, webhook history, allowance window	Reconcile ledger; use audited adjustment rather than duplicate grant.
Published twice	Provider IDs, retries, uncertain timeout and callback ordering	Stop repeated submit; reconcile each accepted publication.
OAuth needs attention	Callback exact origin, encrypted grant, scope and revocation	Reconnect only after configuration/key issue understood.
Scan fails or is empty	SSRF block, robots, crawl stage, rendering gate and limits	Explain the blocked target; do not weaken safe-fetch globally.

11 Backups, rollback and storage migration

Recommended backup scope is PostgreSQL schema/data and migration history, private media and its mapping identities, environment/encryption-key escrow, provider installation metadata and release artifacts. The actual backup provider, region, schedule, retention, restore test, RPO and RTO were not supplied. Set those explicitly before claiming disaster recovery readiness.

Application rollback on Railway should restore the last known healthy deployment or a reviewed revert. Database changes may not be backward compatible; prefer additive migrations and verify the old application can read the new schema before relying on deploy rollback. Never run an improvised reverse migration against a production ledger.

Keep R2 credentials available during rollback so assets created after cutover remain readable. Legacy Supabase objects and migration aliases may still be referenced. Storage migration uses checkpoints, checksums, reference cutover and rollback controls; do not delete old objects merely because copying finished. Retirement requires separate verification and authorization.

A recovery drill should restore into an isolated environment, validate memberships/RLS, reopen representative old and new assets, advance a paused durable job, check financial replay behavior and verify callback URLs cannot reach production accidentally. Record evidence and elapsed time rather than a guessed availability promise.

Implementation sources: docs/storage-migration.md; docs/storage.md; supabase/migrations; docs/operations.md

12 Environment name inventory

Names only. No secret values were read from private .env files. The list combines literal process.env references in non-test TypeScript with keys in .env.example. Template-only entries may be old, optional or read indirectly; they are not proof of active use or required production settings. Dynamic families such as AI_MODEL_<ROUTE>, AI_PRICE_<MODEL> and workspace flag overrides require the family rules in the relevant module.

Variable name	Classification	Evidence
ADMIN_REQUIRE_MFA	Server/integration/runtime configuration	Source: src/server/admin/config.server.ts
AGENTS_DISABLED	Server/integration/runtime configuration	Source: src/server/agents/policy.ts
AI_USER_DAILY_USD	AI/media/routing control	.env.example name only
ALERT_WEBHOOK_URL	Server/integration/runtime configuration	Source: src/server/observability/alerts.ts
APP_DOMAIN	Server/integration/runtime configuration	.env.example name only
APP_URL	Server/integration/runtime configuration	Source: src/server/assets/persist.server.ts
ASSET_METADATA_EXIFTOOL_DIR	Server/integration/runtime configuration	Source: src/server/assets/image-metadata.server.ts
ASSET_METADATA_PYTHON_BIN	Server/integration/runtime configuration	Source: src/server/assets/image-metadata.server.ts
BILLING_ADMIN_EMAILS	Financial/entitlement configuration	Source: src/server/billing/notify.server.ts
BILLING_ADMIN_USER_IDS	Financial/entitlement configuration	Source: src/server/billing/admin.server.ts
BILLING_EMAIL_FROM	Financial/entitlement configuration	Source: src/server/notify/email.server.ts
BILLING_LIVE	Financial/entitlement configuration	.env.example name only
BILLING_PROVIDER	Financial/entitlement configuration	.env.example name only
CANVA_CLIENT_ID	Server/integration/runtime configuration	Source: src/server/connectors/canva/config.server.ts
CANVA_CLIENT_SECRET	Sensitive server configuration	Source: src/server/connectors/canva/config.server.ts
CANVA_TOKEN_ENCRYPTION_KEY	Sensitive server configuration	.env.example name only
CONTACT_FROM_EMAIL	Server/integration/runtime configuration	Source: src/app/(marketing)/api/contact/route.ts
CONTACT_TO_EMAIL	Server/integration/runtime configuration	Source: src/app/(marketing)/api/contact/route.ts

Variable name	Classification	Evidence
CRON_SECRET	Sensitive server configuration	Source: src/server/cron.ts
DISTRIBUTION_PROVIDER	Server/integration/runtime configuration	Source: src/lib/feature-flags.ts
E2E_TEST_EMAIL	Server/integration/runtime configuration	.env.example name only
E2E_TEST_PASSWORD	Sensitive server configuration	.env.example name only
ENV_VALIDATION	Server/integration/runtime configuration	Source: src/instrumentation.ts
FEATURE_FLAG_ASSET_METADATA_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_AUDIENCE_AUTO_SCORE_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_AUDIENCE_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_AUTOPILOT_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_AUTOPILOT_FULL_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_CANVA_MAGIC_LAYERS_ENABLED	Feature gate; inspect owning helper	Source: src/server/connectors/canva/config.server.ts
FEATURE_FLAG_CHAT_TOOLS_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_ENTITLEMENT_OVERRIDES_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_GEO_AGENT_ENABLED	Feature gate; inspect owning helper	Source: src/server/geo/agents/flags.ts
FEATURE_FLAG_GEO_AI_PROBES_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_GEO_RENDERING_ENABLED	Feature gate; inspect owning helper	Source: src/server/geo/render.server.ts
FEATURE_FLAG_GOOGLE_ANALYTICS_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_MARKETING_COACH_WORKFLOW_ENABLED	Feature gate; inspect owning helper	Source: src/server/workflows/marketing-coach-flags.server.ts
FEATURE_FLAG_MCP_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_MEMORY_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_PROOF_ENGINE_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts

Variable name	Classification	Evidence
FEATURE_FLAG_SLACK_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_SOCIALAPI_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_SOCIAL_TRENDS_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_STORIES_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FEATURE_FLAG_UGC_VIDEO_ENABLED	Feature gate; inspect owning helper	Source: src/lib/feature-flags.ts
FFMPEG_PATH	Server/integration/runtime configuration	Source: src/server/guardrails/video-safety.server.ts
FIRECRAWL_API_KEY	Sensitive server configuration	Source: src/lib/firecrawl-gateway.server.ts
FIRECRAWL_BASE_URL	Server/integration/runtime configuration	Source: src/lib/firecrawl-flags.server.ts
FIRECRAWL_MAX_RETRIES	Server/integration/runtime configuration	.env.example name only
FIRECRAWL_NUM_WORKERS_PER_QUEUE	Server/integration/runtime configuration	.env.example name only
FIRECRAWL_TIMEOUT_MS	Server/integration/runtime configuration	.env.example name only
GEO_AGENT_DAILY_RUNS	AI/media/routing control	Source: src/server/geo/agents/service.server.ts
GEO_AGENT_LIVE_PR	AI/media/routing control	.env.example name only
GEO_AGENT_LIVE_REPO	AI/media/routing control	.env.example name only
GEO_AGENT_LIVE_SITE	AI/media/routing control	.env.example name only
GEO_AGENT_MAX_COST_USD	AI/media/routing control	Source: src/server/geo/agents/geo-coding-agent.ts
GEO_CITATION_PROBE_MAX_QUERIES	AI/media/routing control	Source: src/server/geo/citation-probes.server.ts
GEO_FIX_VERIFY_DELAYS	AI/media/routing control	Source: src/server/geo/fixes/verify.server.ts
GEO_PROBE_MAX_QUERIES	AI/media/routing control	Source: src/server/geo/probes.server.ts
GEO_PROBE_MODELS	AI/media/routing control	Source: src/server/ai/task-models.ts
GEO_PROBE_WEB_SEARCH	AI/media/routing control	Source: src/server/geo/tracked-prompts.server.ts
GEO_RENDER_EXECUTABLE	AI/media/routing control	Source: src/server/geo/render.server.ts
GEO_RENDER_NO_SANDBOX	AI/media/routing control	Source: src/server/geo/render.server.ts
GEO_RENDER_WS_ENDPOINT	AI/media/routing control	Source: src/server/geo/render.server.ts
GITHUB_ALLOWED_RETURN_ORIGINS	Server/integration/runtime configuration	Source: src/server/connectors/webflow/config.server.ts

Variable name	Classification	Evidence
GITHUB_API_BASE_OVERRIDE	Server/integration/runtime configuration	Source: src/server/connectors/github/api.server.ts
GITHUB_APP_ID	Server/integration/runtime configuration	.env.example name only
GITHUB_APP_NAME	Server/integration/runtime configuration	.env.example name only
GITHUB_APP_PRIVATE_KEY	Sensitive server configuration	.env.example name only
GITHUB_APP_SLUG	Server/integration/runtime configuration	.env.example name only
GITHUB_CLIENT_ID	Server/integration/runtime configuration	.env.example name only
GITHUB_CLIENT_SECRET	Sensitive server configuration	.env.example name only
GITHUB_INSTALL_VERIFICATION	Server/integration/runtime configuration	.env.example name only
GITHUB_LIVE_REPO	Server/integration/runtime configuration	.env.example name only
GITHUB_WEBHOOK_SECRET	Sensitive server configuration	Source: src/app/api/integrations/github/webhook/route.ts
GIT_COMMIT_SHA	Server/integration/runtime configuration	Source: src/app/api/health/route.ts
GOOGLE_ANALYTICS_CLIENT_ID	Server/integration/runtime configuration	Source: src/lib/feature-flags.ts
GOOGLE_CLIENT_ID	Server/integration/runtime configuration	.env.example name only
GOOGLE_CLIENT_SECRET	Sensitive server configuration	.env.example name only
HELICONE_API_KEY	Sensitive server configuration	Source: src/server/observability/helicone.server.ts
HELICONE_BASE_URL	Server/integration/runtime configuration	Source: src/server/observability/helicone-flags.server.ts
HELICONE_CLICKHOUSE_PASSWORD	Sensitive server configuration	.env.example name only
HELICONE_LOG_PROMPTS	Server/integration/runtime configuration	Source: src/server/observability/helicone-flags.server.ts
HELICONE_MINIO_ROOT_PASSWORD	Sensitive server configuration	.env.example name only
HELICONE_MINIO_ROOT_USER	Server/integration/runtime configuration	.env.example name only
HELICONE_POSTGRES_PASSWORD	Sensitive server configuration	.env.example name only
IMAGE_MODEL_DEFAULT	AI/media/routing control	Source: src/lib/model-router.server.ts
IMAGE_MODEL_EDIT	AI/media/routing control	Source: src/lib/model-router.server.ts

Variable name	Classification	Evidence
IMAGE_MODEL_PREMIUM	AI/media/routing control	Source: src/lib/model-router.server.ts
IMAGE_MODEL_PREMIUM_EDIT	AI/media/routing control	Source: src/lib/model-router.server.ts
KIE_API_KEY	Sensitive server configuration	Source: src/app/api/health/route.ts
KIE_WEBHOOK_HMAC_KEY	Server/integration/runtime configuration	Source: src/app/api/public/hooks/kie/route.ts
LOG_LEVEL	Server/integration/runtime configuration	Source: src/server/observability/logger.ts
MELLOX_ADMIN_USER_IDS	Server/integration/runtime configuration	Source: src/server/billing/admin.server.ts
MELLOX_CREDITS_PER_USD	Financial/entitlement configuration	Source: src/server/links/credits.server.ts
MELLOX_LINK_MARGIN	Financial/entitlement configuration	Source: src/server/links/credits.server.ts
MELLOX_LINK_MAX_ORDER_USD	Financial/entitlement configuration	Source: src/server/links/credits.server.ts
NEXT_PUBLIC_APP_URL	Public build/browser value	Source: src/lib/seo.ts
NEXT_PUBLIC_BING_SITE_VERIFICATIO N	Public build/browser value	Source: src/app/(marketing)/layout.tsx
NEXT_PUBLIC_BOOKING_URL	Public build/browser value	Source: src/app/(marketing)/demo/page.tsx
NEXT_PUBLIC_CLARITY_ID	Public build/browser value	Source: src/marketing/lib/analytics.ts
NEXT_PUBLIC_GA_MEASUREMENT_ID	Public build/browser value	Source: src/marketing/lib/analytics.ts
NEXT_PUBLIC_GOOGLE_CLIENT_ID	Public build/browser value	.env.example name only
NEXT_PUBLIC_GOOGLE_SITE_VERIFICA TION	Public build/browser value	Source: src/app/(marketing)/layout.tsx
NEXT_PUBLIC_LINKEDIN_PARTNER_ID	Public build/browser value	Source: src/marketing/lib/analytics.ts
NEXT_PUBLIC_META_PIXEL_ID	Public build/browser value	Source: src/marketing/lib/analytics.ts
NEXT_PUBLIC_PLAUSIBLE_DOMAIN	Public build/browser value	Source: src/marketing/lib/analytics.ts
NEXT_PUBLIC_SITE_URL	Public build/browser value	Source: src/app/(marketing)/blog/[slug]/page.tsx
NEXT_PUBLIC_SUPABASE_PUBLISHABL E_KEY	Public build/browser value	Source: src/integrations/supabase/client.ts
NEXT_PUBLIC_SUPABASE_URL	Public build/browser value	Source: src/integrations/supabase/client.ts
NEXT_PUBLIC_TURNSTILE_SITE_KEY	Public build/browser value	Source: src/components/auth/AuthCaptcha.tsx
NEXT_PUBLIC_VERCEL_ANALYTICS	Public build/browser value	Source: src/marketing/lib/analytics.ts
NEXT_PUBLIC_VERCEL_ENV	Public build/browser value	Source: src/marketing/lib/analytics.ts

Variable name	Classification	Evidence
NEXT_RUNTIME	Server/integration/runtime configuration	Source: src/instrumentation.ts
NODE_ENV	Server/integration/runtime configuration	Source: src/app/admin-lab/page.tsx
NOTION_CLIENT_ID	Server/integration/runtime configuration	Source: src/server/connectors/notion/config.server.ts
NOTION_CLIENT_SECRET	Sensitive server configuration	Source: src/server/connectors/notion/config.server.ts
NOTION_REDIRECT_URI	Server/integration/runtime configuration	Source: src/server/connectors/notion/config.server.ts
NOTION_TOKEN_ENCRYPTION_KEY	Sensitive server configuration	.env.example name only
OPENROUTER_API_KEY	Sensitive server configuration	Source: src/app/api/health/route.ts
OPENROUTER_WEBHOOK_SECRET	Sensitive server configuration	Source: src/app/api/public/hooks/openrouter-video/route.ts
PATH	Server/integration/runtime configuration	Source: src/server/assets/image-metadata.server.ts
PEXELS_API_KEY	Sensitive server configuration	Source: src/app/api/pexels/video/route.ts
PLAN_AGENCY_GEO_MAX_PAGES	Server/integration/runtime configuration	.env.example name only
PLAN_GROWTH_GEO_MAX_PAGES	Server/integration/runtime configuration	.env.example name only
PLAN_STARTER_DAILY_USD	Server/integration/runtime configuration	.env.example name only
PLAN_STARTER_GEO_MAX_PAGES	Server/integration/runtime configuration	.env.example name only
PLAN_STARTER_MONTHLY_POSTS	Server/integration/runtime configuration	.env.example name only
PLAN_STARTER_MONTHLY_USD	Server/integration/runtime configuration	.env.example name only
POST_FOR_ME_API_KEY	Sensitive server configuration	Source: src/lib/feature-flags.ts
POST_FOR_ME_WEBHOOK_SECRET	Sensitive server configuration	Source: src/app/api/public/hooks/postforme/route.ts
R2_ACCESS_KEY_ID	Server/integration/runtime configuration	Source: src/server/assets/storage.server.ts
R2_ACCOUNT_ID	Server/integration/runtime configuration	Source: src/server/assets/storage.server.ts
R2_BUCKET_NAME	Server/integration/runtime configuration	Source: src/server/assets/storage.server.ts
R2_ENDPOINT	Server/integration/runtime configuration	Source: src/server/assets/storage.server.ts

Variable name	Classification	Evidence
R2_SECRET_ACCESS_KEY	Sensitive server configuration	Source: src/server/assets/storage.server.ts
RAILWAY_GIT_BRANCH	Server/integration/runtime configuration	Source: src/app/api/health/route.ts
RAILWAY_GIT_COMMIT_SHA	Server/integration/runtime configuration	Source: src/app/api/health/route.ts
REDIS_URL	Server/integration/runtime configuration	Source: src/app/api/health/ready/route.ts
RESEND_API_KEY	Sensitive server configuration	Source: src/app/(marketing)/api/contact/route.ts
RIXOT_API_KEY	Sensitive server configuration	Source: src/server/links/rixot/client.server.ts
RIXOT_BASE_URL	Server/integration/runtime configuration	Source: src/server/links/rixot/client.server.ts
SDR_ADMIN_TOKEN	Sensitive server configuration	Source: src/app/api/public/hooks/sdr-reconcile/route.ts
SDR_BASE_URL	Server/integration/runtime configuration	Source: src/app/api/public/hooks/sdr-reconcile/route.ts
SDR_WEBHOOK_BASE_URL	Server/integration/runtime configuration	Source: src/lib/sdr-provisioning.server.ts
SDR_WEBHOOK_TOLERANCE_SECONDS	Server/integration/runtime configuration	Source: src/app/api/public/hooks/sdr/route.ts
SENTRY_DSN	Server/integration/runtime configuration	Source: src/server/observability/errors.ts
SHARE_LINK_ENCRYPTION_KEY	Server/integration/runtime configuration	.env.example name only
SLACK_CLIENT_ID	Server/integration/runtime configuration	Source: src/server/slack/security.server.ts
SLACK_CLIENT_SECRET	Sensitive server configuration	Source: src/server/slack/security.server.ts
SLACK_SIGNING_SECRET	Sensitive server configuration	Source: src/server/slack/security.server.ts
SLACK_TOKEN_ENCRYPTION_KEY	Sensitive server configuration	.env.example name only
SOCIALAPI_API_KEY	Sensitive server configuration	Source: src/lib/feature-flags.ts
SOCIALAPI_BASE_URL	Server/integration/runtime configuration	Source: src/lib/socialapi/workspace.server.ts
STRIPE_SECRET_KEY	Sensitive server configuration	Source: src/server/billing/stripe-account.server.ts
STRIPE_WEBHOOK_SECRET	Sensitive server configuration	Source: src/server/billing/stripe-account.server.ts
SUPABASE_PUBLISHABLE_KEY	Server/integration/runtime configuration	Source: src/integrations/supabase/client.ts
SUPABASE_SERVICE_ROLE_KEY	Server/integration/runtime configuration	Source: src/integrations/supabase/client.server.ts

Variable name	Classification	Evidence
SUPABASE_URL	Server/integration/runtime configuration	Source: src/app/.well-known/oauth-protected-resource/[[...path]]/route.ts
TAVILY_API_KEY	Sensitive server configuration	Source: src/lib/tavily-flags.server.ts
TAVILY_BASE_URL	Server/integration/runtime configuration	Source: src/lib/tavily-gateway.server.ts
TAVILY_MAX_RETRIES	Server/integration/runtime configuration	.env.example name only
TAVILY_TIMEOUT_MS	Server/integration/runtime configuration	.env.example name only
TRIGGER_API_URL	Server/integration/runtime configuration	Source: src/server/trigger/flags.server.ts
TRIGGER_PROJECT_REF	Server/integration/runtime configuration	.env.example name only
TRIGGER_SECRET_KEY	Sensitive server configuration	Source: src/server/trigger/flags.server.ts
UGC_DEFAULT_MODEL	AI/media/routing control	Source: src/server/ugc/models.server.ts
UGC_MAX_CONCURRENT_RENDERS	AI/media/routing control	Source: src/server/ugc/service.server.ts
UNSPLASH_ACCESS_KEY	Server/integration/runtime configuration	Source: src/app/api/unsplash/random/route.ts
VIDEO_PROVIDER	AI/media/routing control	Source: src/server/ugc/models.server.ts
VIDEO_PROVIDER_CINEMATIC	AI/media/routing control	.env.example name only
VIDEO_PROVIDER_DRAFT	AI/media/routing control	.env.example name only
VIDEO_PROVIDER_FALLBACK	AI/media/routing control	Source: src/server/ugc/models.server.ts
VIDEO_PROVIDER_LONG	AI/media/routing control	.env.example name only
VIDEO_PROVIDER_PREMIUM	AI/media/routing control	.env.example name only
VIDEO_PROVIDER_STANDARD	AI/media/routing control	.env.example name only
VIDEO_PROVIDER_VARIATION	AI/media/routing control	.env.example name only
VISUAL_CAPTURE_BROWSER_CHANNEL	Server/integration/runtime configuration	Source: src/server/brand-visuals/browser.server.ts
VISUAL_CAPTURE_BROWSER_PATH	Server/integration/runtime configuration	Source: src/server/brand-visuals/browser.server.ts
WEBFLOW_CLIENT_ID	Server/integration/runtime configuration	.env.example name only
WEBFLOW_CLIENT_SECRET	Sensitive server configuration	.env.example name only
WEBFLOW_REDIRECT_URI	Server/integration/runtime configuration	.env.example name only
WEBFLOW_TOKEN_ENCRYPTION_KEY	Sensitive server configuration	.env.example name only

Variable name	Classification	Evidence
WORDPRESS_CLIENT_ID	Server/integration/runtime configuration	.env.example name only
WORDPRESS_CLIENT_SECRET	Sensitive server configuration	.env.example name only
WORDPRESS_REDIRECT_URI	Server/integration/runtime configuration	.env.example name only
WORDPRESS_TOKEN_ENCRYPTION_KEY	Sensitive server configuration	Source: src/server/fns/connectors.ts

Implementation sources: .env.example; src/server/env.ts

13 Operating records still needed

• Railway service inventory, regions, replicas, resource budgets and owner/escalation contacts.

• Applied database migrations, scheduled cron jobs and release-to-schema compatibility record.

• Configured provider/model overrides, app approvals, scopes and funded vendor accounts.

• Invoice-backed cost report, operational spend caps and alert thresholds.

• Backup/restore evidence, retention/export/deletion policy, RPO/RTO and contractual SLA.

• Documented production smoke results for auth, R2, payments, generation, social delivery and website writes.

14 Source ledger and verification limits

Repository-relative paths below identify the owning implementation. The companion source folder includes an extracted inventory and SHA-256 hashes for source files and migrations. The inventory is a static extraction, not a database dump or a successful production test. Generated database types describe their checked-in state; migrations define actual SQL constraints, defaults and policies.

The Mellox connector failed with an internal error. The public site could not be fetched by the web tool or local network, and browser initialization was unavailable. Consequently public-site rendering, deployed commit, enabled credentials, vendor approvals, production schema, invoices, real usage and exact Railway service topology were not independently inspected. The local marketing implementation was used as a secondary source. No live provider generation or publication was triggered.

Source	SHA-256 prefix
Dockerfile	430225c9b5bc79c7
next.config.ts	bc5ebc3b26f710f9
src/server/env.ts	a814c54a60a11cb0
docs/deployment-guide.md	a607a06aa6bf0a3e
src/app/health/route.ts	a748db71cb16ffa1
src/app/api/health/route.ts	4b8000f7cc687574
src/app/api/health/ready/route.ts	6b771bafa6db6ca5
scripts/start-production.mjs	29457c96d4aa912a
src/instrumentation.ts	5e3caad36277cc50
src/server/cron.ts	a6ba76d8d76537e4
supabase/migrations	directory reference
docs/storage.md	a019ec513b248c76
src/server/assets/storage.server.ts	5f20083ebbddc447
src/server/connectors	directory reference
src/server/billing/readiness.server.ts	21fce896eb96ab56
src/lib/feature-flags.ts	e56ac35e17629593
src/server/platform/flags.server.ts	30ef38a324aada31
src/server/platform/switches.server.ts	dcd79c367132d502
src/server/platform/spend-caps.server.ts	c5e4e866004d91f2
src/server/platform/routing-overrides.server.ts	64efb98109da4141
src/app/api/public/hooks	directory reference
src/app/api/public/hooks/run-schedules/route.ts	8f479418bbf8ff6e

Source	SHA-256 prefix
src/server/route.ts	391e0eb6fa6c43a7
src/server/safe-fetch.ts	9f217ee77851db86
src/server/guardrails	directory reference
src/server/mcp	directory reference
src/server/admin/route.ts	e04b317035a7a470
src/integrations/supabase/client.ts	3d76962d05c21248
docs/security.md	5fda9dc384550f40
docs/privacy.md	2ce0e7e923584f31
docs/self-hosted-integrations.md	20fc25ab6c39c75d
src/server/observability	directory reference
docs/operations.md	f48e2d659373decf
docs/storage-migration.md	890c6f55243cb487
.env.example	00e8b119b49057d4